



Society of Petroleum Engineers

SPE-227075-MS

First Implementation of Dry Friction Reducer Using Fit-For-Purpose Unit in the Middle East

R. Tineo, M. AlQusom, and O. Bawareth, SLB, Dhahran, Saudi Arabia

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227075-MS](https://doi.org/10.2118/227075-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Slickwater is the most commonly used fracturing fluid system for completing wells in unconventional reservoirs. Traditionally, the friction reducer used to formulate slickwater is a polyacrylamide transported in liquid form, either as a slurry or an emulsion mixture, to facilitate fluid hydration and wellsite delivery using standard equipment. The objective of this work is to implement a dry friction reducer additive that meets the fluid requirements, and it is injected with a fit-for-purpose unit, ensuring flawless operations.

The first step consisted of laboratory work conducted to screen several dry friction reducer options and select materials that could deliver adequate hydration time, drag reduction, viscosity, and regained permeability. Secondly, a fit-for-purpose skid, specifically designed to inject friction reducers in powder form, was deployed for the first time in the Middle East. This new unit allows for the automatic control of the desired friction reducer concentration, reduces the operational footprint, and mitigates dust-related health and environmental risks. A series of sensors are installed to closely monitor the product delivery. Lastly, a yard test is conducted to calibrate the unit with the actual material selected from the laboratory screening process.

Laboratory results indicated that the selected dry friction reducer offers a drag reduction of around 70%, regained permeability close to 100%, and viscosities between 3 cp and 30 cp at 511 sec^{-1} at ambient temperature, using concentrations from 3.5 ppt to 35 ppt. Complete hydration was observed within 60 seconds.

Two horizontal wells were completed using the selected dry friction reducer and the fit-for-purpose skid. A total of ninety-two hydraulic fracturing stages were pumped as per design at a rate above ninety-five barrels per minute. The average tubular friction observed during the operation was similar to or slightly lower than stages pumped using legacy friction reducers in liquid form in offset wells under similar conditions. Physical material balance demonstrated that the unit precisely delivered the product, with a maximum deviation of within one percent.

It is proven that dry friction reducers can be effectively implemented at a large scale for unconventional operations in the Middle East.

This innovative solution can benefit operators and service companies by reducing material and transportation logistics, CO₂ emissions, and storage requirements while maintaining a flawless operation delivery.

Introduction

The stimulation process of hydraulically fracturing a formation involves injecting a volume of fluid to break the formation down to increase the connectivity with the reservoir which consequently will increase the conductivity and productivity of the reservoir (Kelvin Abaa 2021). This is a popular process in the oil and gas industry, especially for reservoirs with low permeability. Several frac fluids since the beginning of the hydraulic fracturing process had been used in the Permian Basin, which was called "carrying agents". Those agents were classified as: (1) hydrocarbon gels, (2) aqueous gels, (3) emulsions, (4) refined oil, (5) lease oil, and (6) miscellaneous fluids (R. E. HURST, et al. 1954).

With the rapid advancement of the unconventional resources and the increasing demand of natural gas at relatively low prices, service provides introduced new solutions to lower the well completion cost (P.S. Patel, et al. 2014) (P.S. Patel, et al. 2014). One of them is using effective low cost frac fluid that will have the physical properties to successfully place the designed frac job while minimizing the damage being introduced to the formation. Since then, slickwater has been widely used in all frac operations worldwide. Slickwater frac fluid consists mainly of friction reducer, to reduce drag and consequently reduce treating pressure allowing pumping the job at high rate, with some other additives.

Liquid friction reducers were commonly pumped due to their operational simplicity. The next generation of friction reducers were introduced by several manufacturers to be high viscose friction reducers, which is a polyacrylamide based chemical able to provide a significant viscosity and drag reduction at lower concentration compared to the regular friction reducer (Nancy Zakhour, et al. 2021). However, using liquid fiction reducer comes with its own challenges i.e. logistics, HSE, and high cost since it comes in liquid format, generally carrying fluid varies from mineral oil to gasoline product that can be easily used to dilute friction reducer polymer in them. With the increasing pressure on low completion cost and greener solutions dry friction reduces rose as key solution to lower the cost of completing well and improving environmental impact.

Incorporating dry FR and HVFR revealed inefficiencies in the existing legacy guar hydration equipment's ability to pump these products effectively. Although modifications to the legacy equipment enhanced the process, shortcomings in the product deployment technology persisted. The procedure remained labor-intensive and lacked the necessary agility and control to meet current stimulation demands. (C. Anderson 2023)

In this paper we present the first application of the dry friction reducer in MENA along with fit-for-purpose delivery system. Will go through the screening process for the products and the trail tests done before the main deployment in the field

Friction reducer screening

The screening process for dry-HVFR (High Viscosity Friction Reducer) products was designed to evaluate their performance against key criteria to ensure their suitability for field execution while also incorporating the cost aspect in the screening process. Figure 1 summarizes the screening process that each product went through to be shortlisted for the field trial.

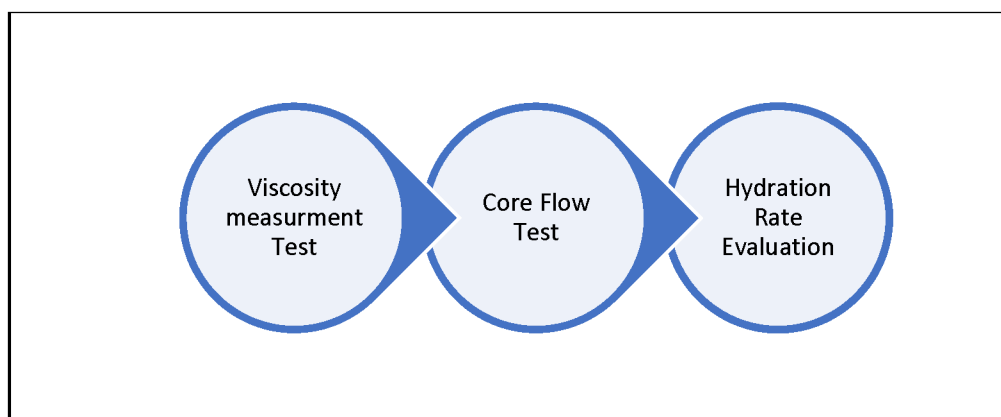


Figure 1—Products Screening Workflow

A total of seven products were subjected to a rigorous series of laboratory tests, which included viscosity measurement, regained permeability, and hydration time evaluation.

Viscosity Measurement

The primary focus of this test was to determine whether the viscosity generated by each product met the threshold required to ensure adequate sand-carrying capacity during job execution. This step is crucial to ensure proper proppant transport and placement into the formation.

The same source of water was used in all tests to eliminate any source of error.

After examining viscosity profiles for all seven products as shown in Figure 2 & Figure 3, products B and E were eliminated since they showed lower than required viscosity values for most of the concentrations. Additionally, product J showed extremely low viscosity values, specifically between concentrations 20-26 ppt which led to its elimination.

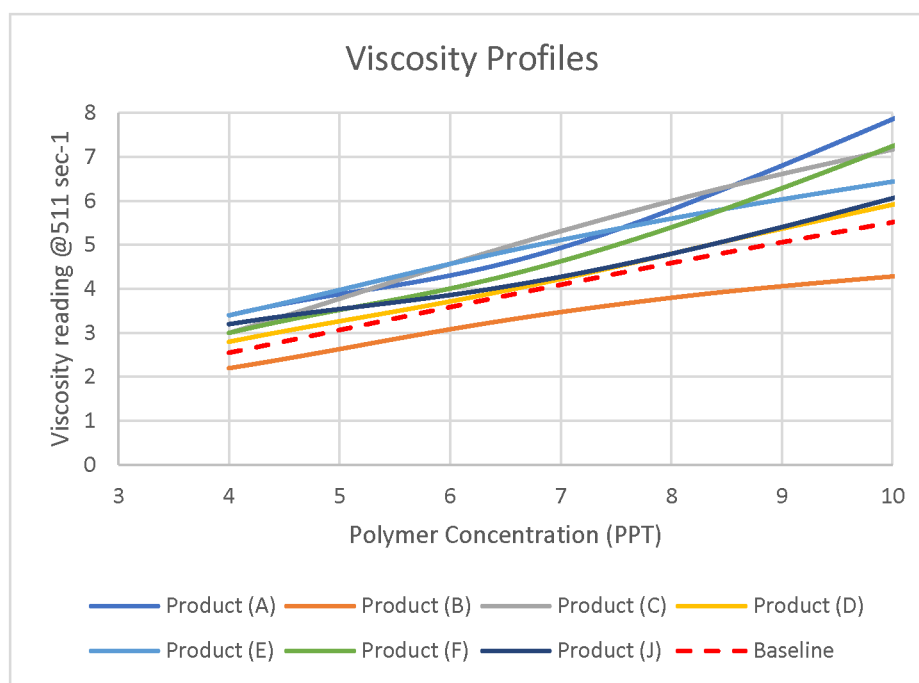


Figure 2—Zoomed-in viscosity profile for all products at different concentrations.

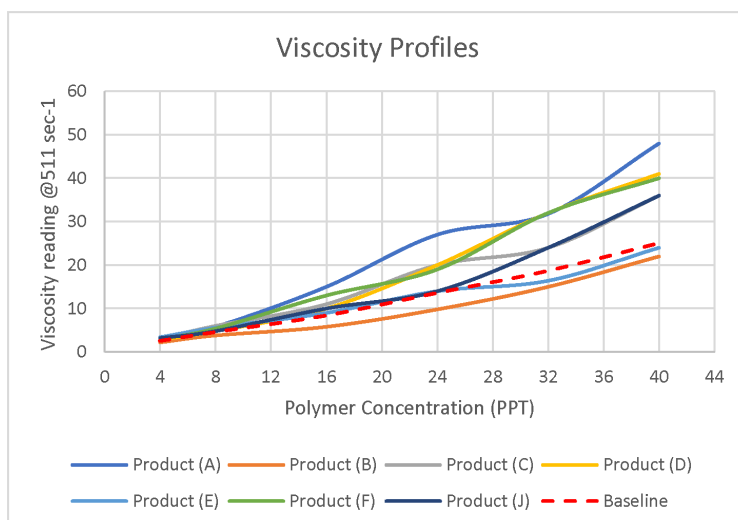


Figure 3—Viscosity Profiles for all products at different concentrations

Core Flow Test

This test was conducted to quantify the degree of formation damage introduced by each product. Maintaining formation permeability is critical for ensuring optimal post-treatment productivity, and this test helped identify products with minimal impact on the reservoir's natural flow characteristics.

The test was formulated to simulate actual stimulation job, in which brine will be injected into the core at different rate to calculate initial permeability and then inject a specified volume of frac fluid in the opposite direction at a specific rate. Following that an 18-hour shut-in period before resume injecting brine to calculate final permeability, from which will quantify the damage introduced to the formation comparing the initial and final permeabilities.

Four products passed viscosity test and core flow tests were done as shown in Figure 4 to Figure 8. Based on the set threshold for the allowable damage to be introduced to the formation, products C, D and F met the requirement. Whereas product A failed to meet minimum requirement in that regard.

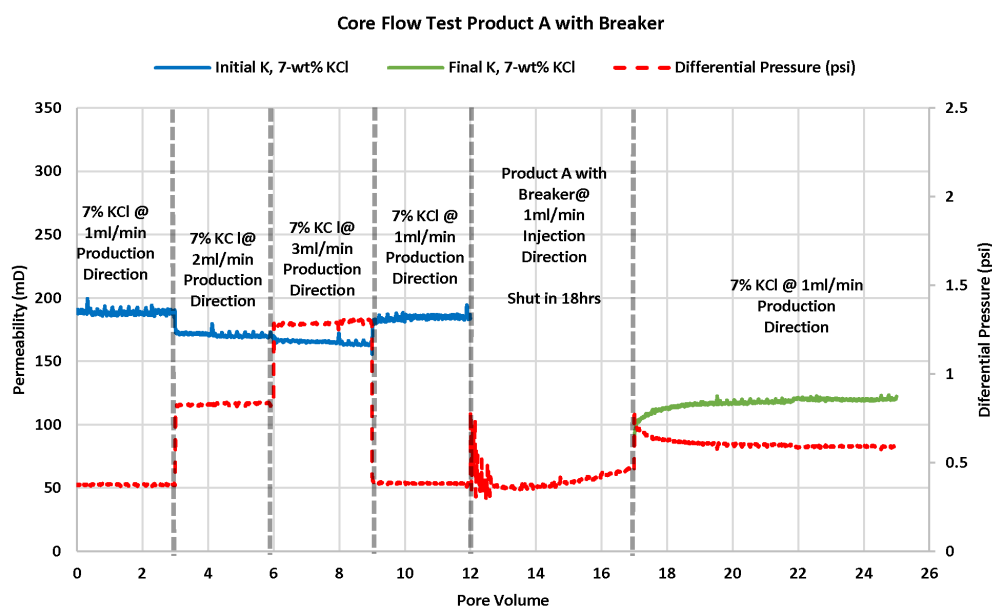


Figure 4—Core Flow test for Product A with Breaker

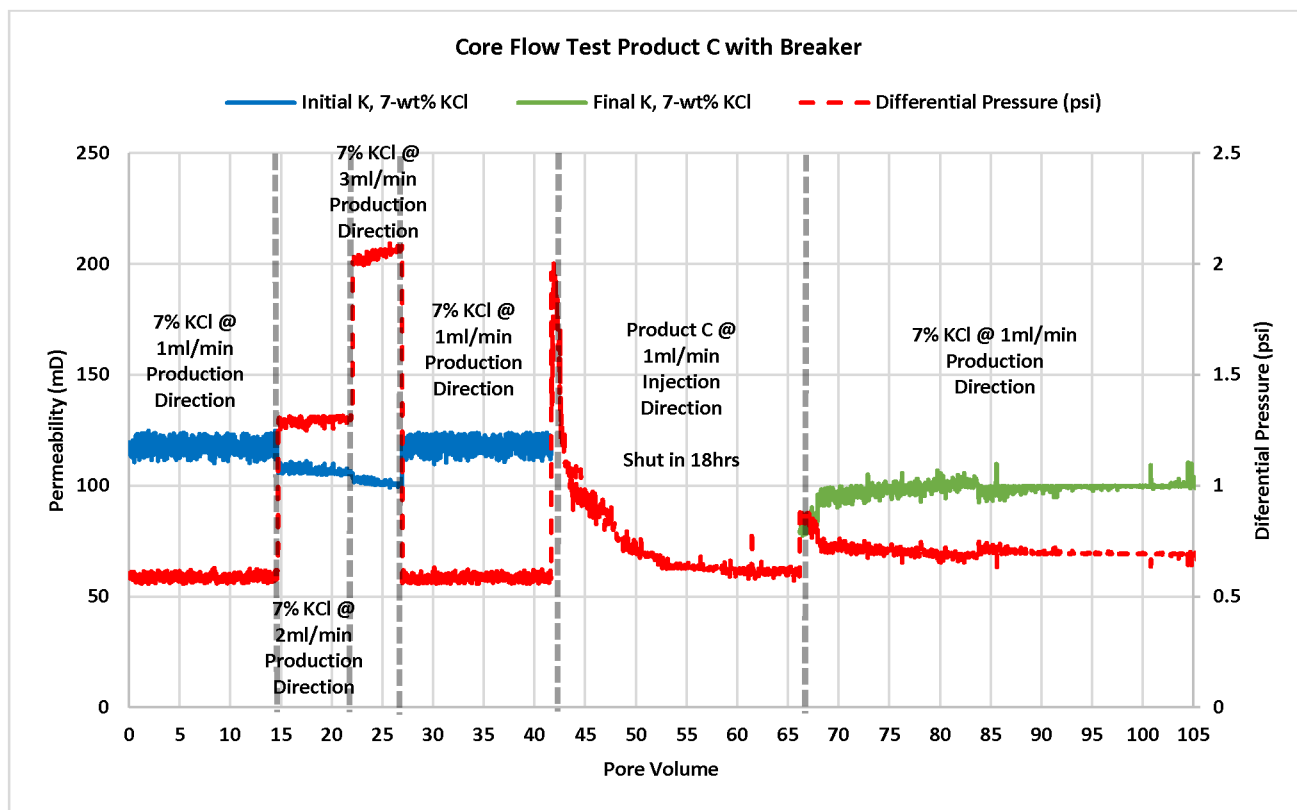


Figure 5—Core Flow Test Product C with Breaker

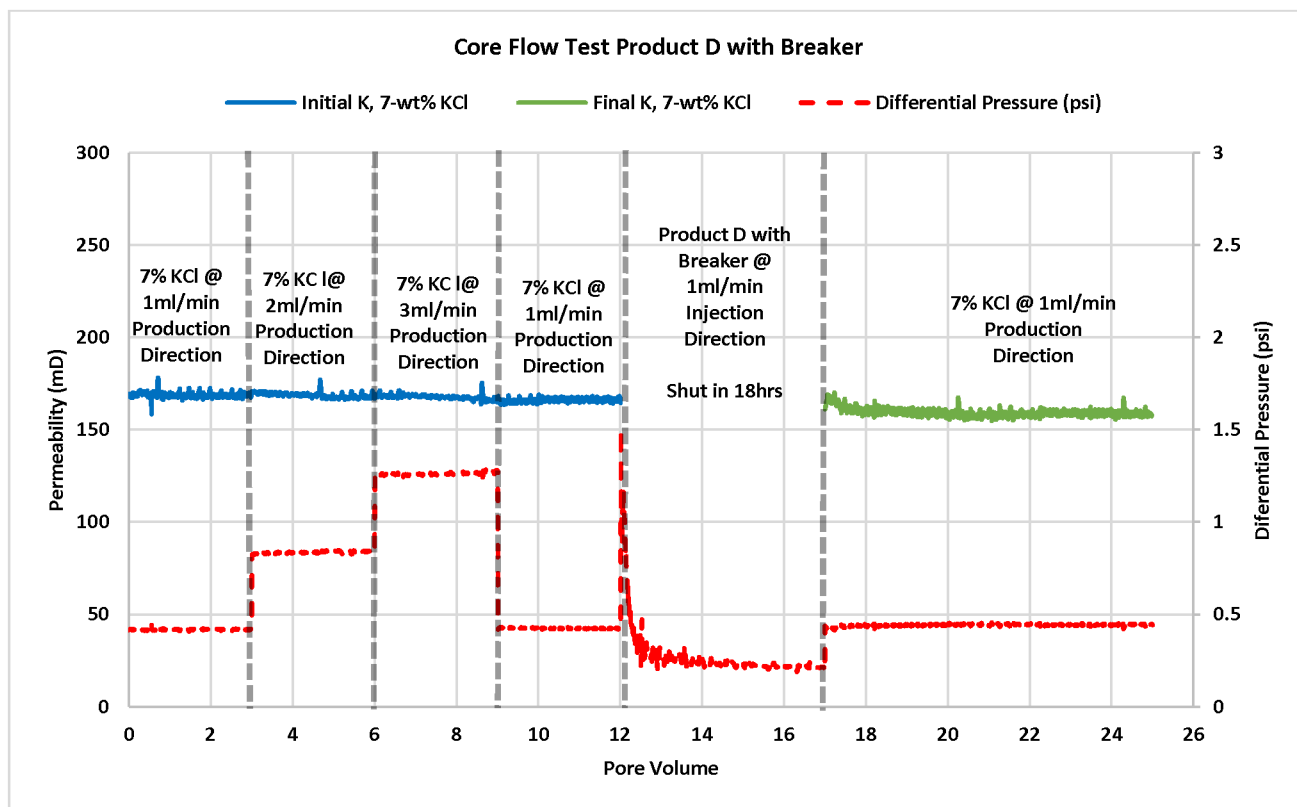


Figure 6—Core Flow Test Product D with Breaker

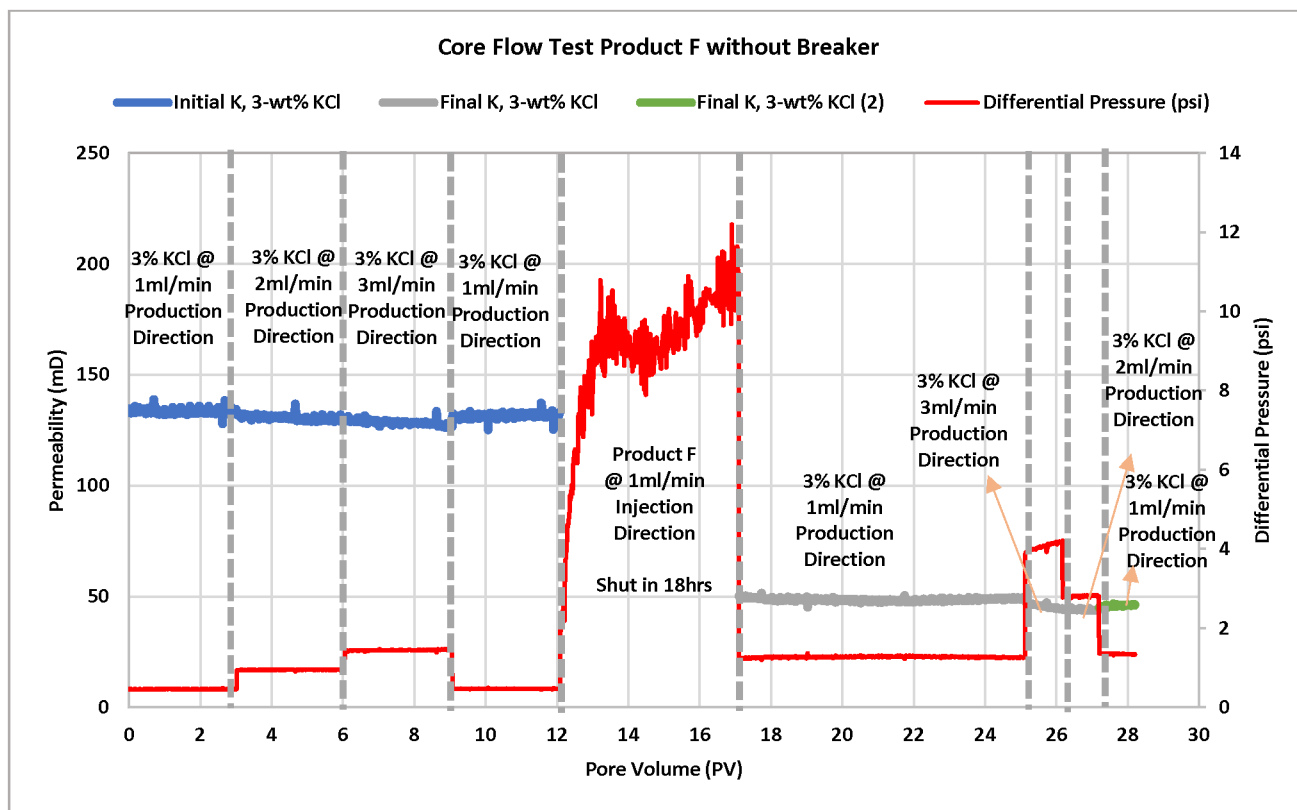


Figure 7—Core Flow Test Product F without Breaker

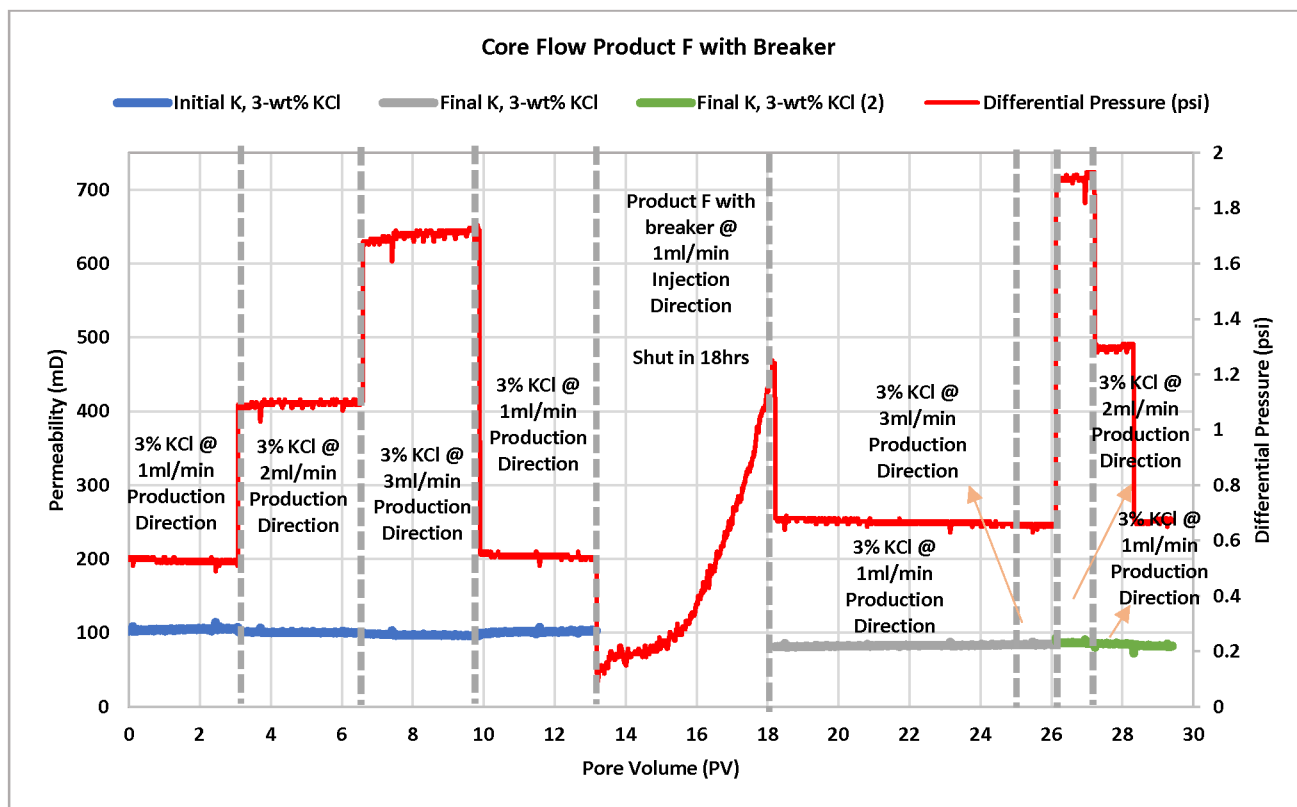


Figure 8—Core Flow Product F with Breaker

Hydration Rate Evaluation

The hydration test assessed how quickly each product achieved its full carrying capacity. Rapid hydration is essential to align with the pumping time required during field execution, ensuring that the product reaches its functional viscosity within operational constraints.

Product C& F were the two products that made it to the hydration test. Based on the hydration test below, both products reached the required viscosity within the required time

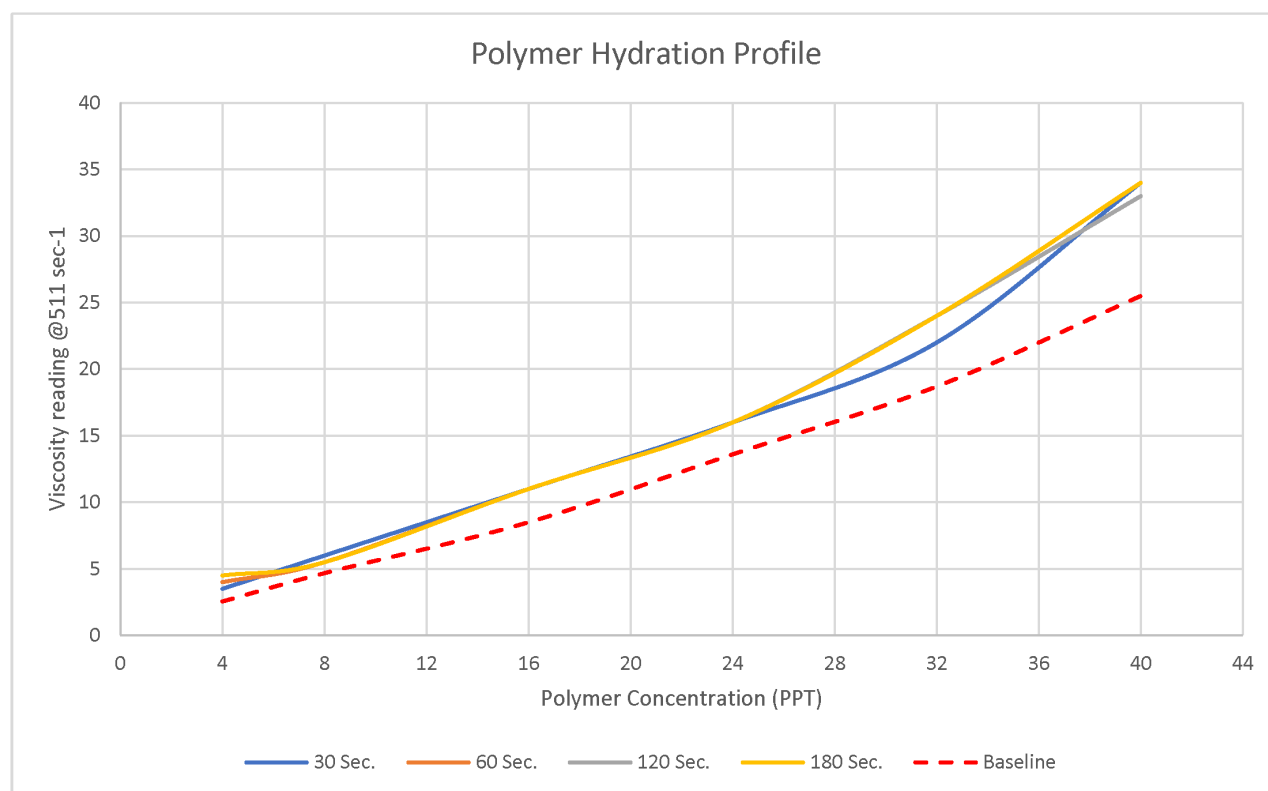


Figure 9—Hydration profile for product F

After completing all the tests, only three of the eight products met the established criteria for viscosity, regained permeability, and hydration time. These three products demonstrated the required performance characteristics to ensure effective proppant transport, minimal formation damage, and operational efficiency during the job execution.

Fit for purpose Unit

Figure 11 shows the fit-for-purpose unit skid used in this project. It is a modular solution designed for metering, hydrating, and injecting dry friction reducer in precise concentrations. It is a compact unit with a small footprint of only 6.5 ft in width and 8.5 ft in length, a nominal height of 5.7 ft, and 16.4 ft including the material bins. It is electrically powered, thus reducing CO₂ emissions.



Figure 10—Polybin details



Figure 11—Modular fit-for-purpose skid

The dry HVFR is transported and delivered to the location in sealed polybin cartridges that can hold up to 3,500 lb [Figure 10](#). These cartridges are loaded on top of the unit and allow for dust-free operation. Two independent bins are mounted on the unit at any given time, providing redundancy and the ability to perform hot-swapping of bins for continuous operations.

The unit delivers up to 50 lb/min of powder rate and a slurry rate of 8 bpm with 100 psi discharge pressure, delivering the dry HVFR up to 150 ppt at the unit output. When product delivery requirements exceed 50 lb/min, additional units can be rigged up to achieve the desired rate.

An automated control system allows for unmanned remote operations. It receives the treatment clean fluid rate and dry product setpoints from the frac van to control the solid-to-liquid ratio. The water stream

is established through an onboard centrifugal pump, and the dry material is automatically metered and conveyed using augers into a flush bowl where the material is hydrated [Figure 12](#).

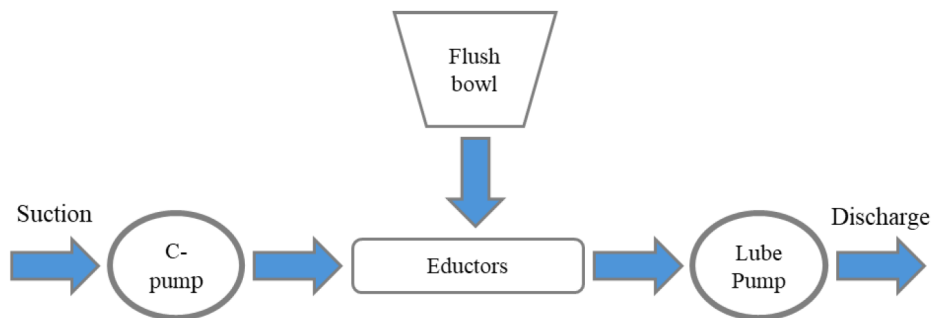


Figure 12—Flow- blending path

Yard Test

As part of the Dry-HVFR delivery system commissioning and qualification process, a yard test representative of the field split-stream setup, but on a smaller scale, was conducted. [Figure 15](#) illustrates the rig up layout, which included the following components on the low-pressure side: four frac tanks, a centrifugal pump, a fracturing blender, and the dry-HVFR skid. On the high-pressure side, there were three frac pumps and a three-inch treating line. A high-pressure to low-pressure manifold was also included.

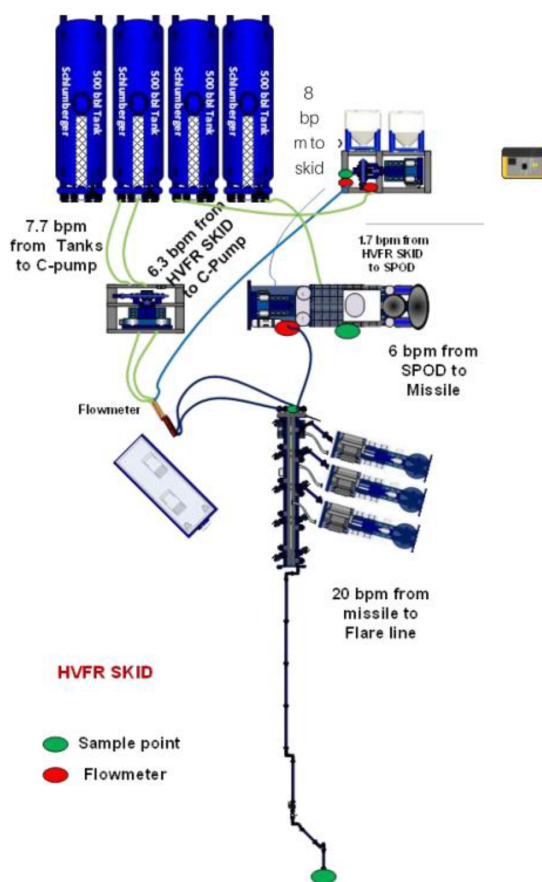


Figure 13—Yard Test Layout

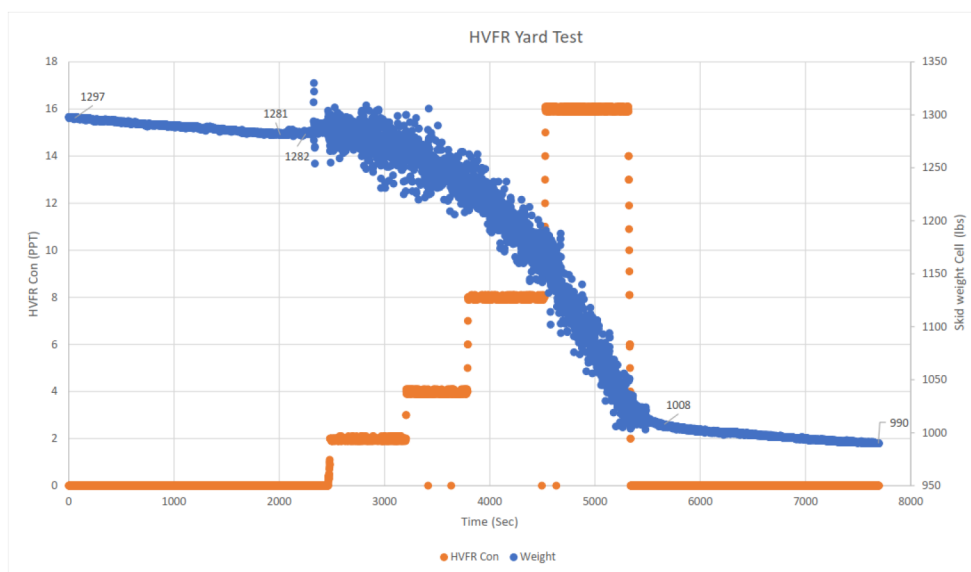


Figure 14—Overall test summary

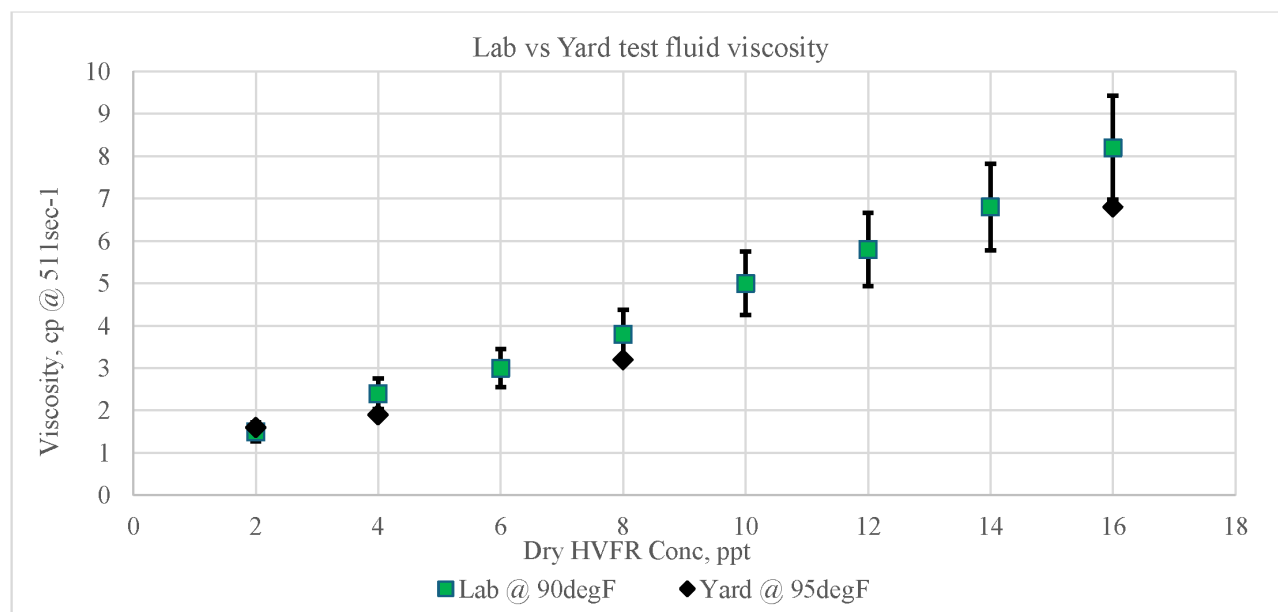


Figure 15—Lab vs Yard test viscosity

Four fluid sampling points were established: one at the skid output, another at the blender discharge, another at the manifold suction, and one at the three-inch treating line output to simulate the fluid entering the wellhead. Fluid viscosity and mass balance measurements were used to validate that the unit was delivering the product properly at the required concentration.

A known volume of fluid was pumped at a constant concentration, and then the total calculated product used was validated against the actual mass consumed based on weight cell readings, as illustrated in Table 1. A deviation of 2% was observed in the water volume when comparing the flowmeter readings against the water straps, and a deviation of 3.4% was observed in the dry-HVFR mass when comparing the unit totalizer and the actual weight difference between initial and final readings.

Table 1—HVFR Skid constant concentration deliverability test results

	Water, bbl	Dry HVFR, lb
Initial Strap	1915	1504
Final Strap	1405	1335
Actual	510	169
Unit Totalizer	520	174.7
%deviation	2.0%	3.4%

Additionally, a multi-concentration viscosity deliverability was conducted varying the dry-hvfr concentration continuously in steps, 2ppt, 4ppt, 8ppt and 16 ppt and measuring the fluid viscosity at each step as illustrated in Figure 14. Table 2 shows the material delivery per step, in general, very small deviations were observed. The 2ppt step showed 12.8% deviation, but it is important to notice that only 23.35 lbs were consumed in that step.

Table 2—Overall test mass balance

Step	Material consumption, lb		
	Weight Cell	Unit Totalizer	Deviation
2 PPT	23.35	20.37	12.8%
4 PPT	33	32.93	0.2%
8 PPT	80.8	82.12	1.6%
16 PPT	165	170.4	3.3%
Total	302.2	305.8	1.2%

Table 3 presents the final viscosity measurements recorded at the wellhead, providing critical data for assessing the fluid's performance under operational conditions. Figure 15 illustrates a comparison between the viscosity values obtained from lab tests and those from yard tests. Notably, the yard test results fall within a 15% margin of the lab test values, indicating a close alignment between experimental and practical outcomes

Table 3—Viscosity measurements

Step	Viscosity, cP	Temp, degF
2 PPT	1.6	95
4 PPT	1.9	95
8 PPT	3.2	95
16 PPT	6.8	95

Case Study

Operational set up

The operational setup for Dry HVFR deployment focuses on controlled polymer delivery, field adaptability, and maintaining product integrity under operational shear. The Hydration skid is positioned upstream of the blender, allowing the dry polymer to be introduced ahead of the blender. While this setup exposes the polymer to shear through both the blender and high-pressure pumps, it is acceptable for slickwater and moderate-viscosity applications, provided dosing rates and shear sensitivity are managed properly.

Prior to operations, the dry polymer is stored in sealed bins to prevent moisture exposure. Powder sensor calibration on the hydration skid is a mandatory pre-job step. It involves an empty adjustment, a full powder-adjustment using sample material, and a function test — all executed using a magnet-triggered sequence for each sensor on Bin 1 and Bin 2. This ensures the system accurately detects powder flow and prevents false readings during operation.

Water analysis and on-site viscosity checks with job-specific water are used to establish baselines. During pumping, auger RPM is monitored and adjusted to match the designed downhole concentration. Visual confirmation of product flow, torque trend monitoring, and powder level tracking are used to detect and resolve issues such as ratholing or bridging in real time.

The setup is optimized for Split Stream Operations (SSO), ensuring that the polymer is hydrated in a blended stream before reaching the fracturing pumps. This controlled hydration step promotes consistent fluid behavior and helps maintain the expected wear profile of fluid end components by avoiding the passage of unhydrated solids through high-pressure sections. The design also simplifies the fluid system layout, reduces chemical handling exposure, and enables full compatibility with existing frac infrastructure. The modular nature of the hydration skid supports rapid rig-up and rig-down, making it suitable for multi-well pad operations and tight stage schedules.

Treatment parameters comparison

Overall, dry-HVFR showed excellent results in comparison with liquid-HVFR. An averaged of 3% reduction in friction was observed in wells utilizing dry-HVFR with 9% less product used compared to liquid-HVFR, considering a 4:1 ration dry to liquid concentration.

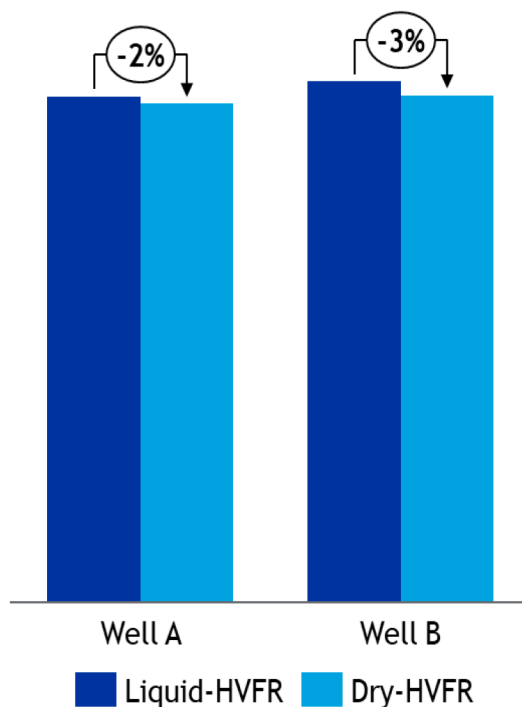


Figure 16—Average Friction Pressure Comparison between wells pumped with liquid-HVFR and wells pumped with Dry-HVFR

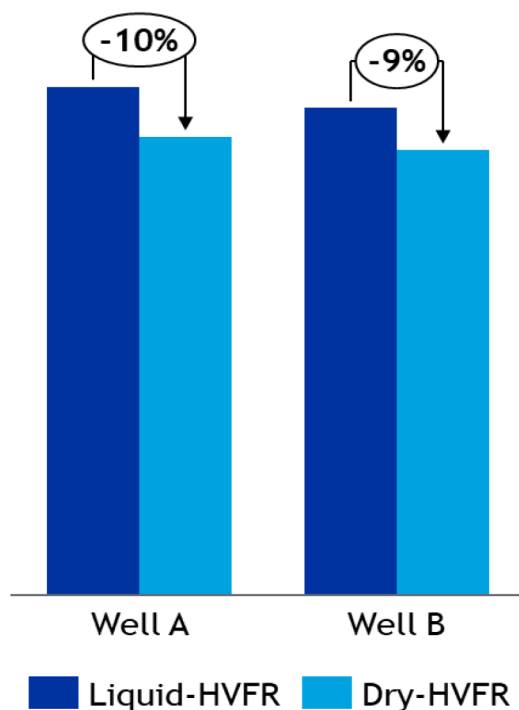


Figure 17—Friction reducer average concentration for each well

Material Delivery results

For the material deliver, four different methods were used to track material consumption:

1. Execution raw data
2. Weight cell on the skid
3. Physical Bin Strap
4. Field scale

Across all measurement methods, the skid delivered product for each well with an accuracy exceeding 98%, Figure 18.

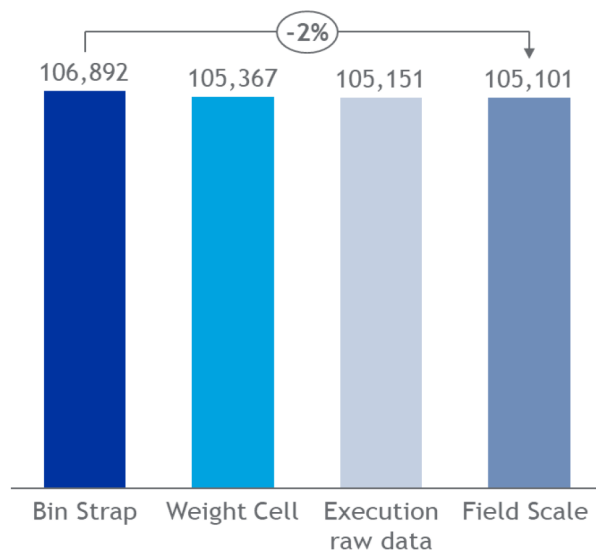


Figure 18—Product consumed based on different measuring methods

Conclusions

The study successfully demonstrates the efficacy of the selected dry friction reducer and delivery system in enhancing hydraulic fracturing operations in unconventional wells within the Middle East. Laboratory tests confirmed that the friction reducer delivers a drag reduction of approximately 70% and achieves permeability regain close to 100%. The viscosity range between 3 cp and 30 cp at 511 sec⁻¹ at ambient temperature was achieved using concentrations from 3.5 ppt to 35 ppt, with complete hydration occurring within 60 seconds. Yard tests further validated the performance, showing that the unit delivered the dry product with an accuracy of one percent. In practice, the application of the dry friction reducer and the fit-for-purpose skid was successfully implemented in two horizontal wells, completing a total of ninety-two hydraulic fracturing stages at rates exceeding ninety-five barrels per minute. The observed friction pressures during these operations were comparable to or slightly lower than those recorded in offset wells using legacy liquid friction reducers under similar conditions. Moreover, the physical material balance indicated precise product delivery, with a maximum deviation of within two percent. These results confirm that dry friction reducers can be effectively deployed on a large scale for unconventional operations, offering a promising alternative to traditional liquid friction reducers in the region.

Acknowledgment

We extend our sincere gratitude to SLB for their invaluable support and for granting permission to publish this paper, specially to the following individuals: Christine Brown, Sean McCallum, Luis Gonzalez, Ayidh Alqahtani, Naif Marzooq Alotaibi, Nawaf Aldhafeeri, Ali Baradwan, Mujtaba Abdullah Alabdulmohsen, Ali Alhussain, Ahmed Mohsen Al-Hejji, Ali Radhi Boareesh, Ibrahim Abdullah Al Shomli, Abdullah Aljaber, Yaqob Younis Alatiyah, Ghadeer Mohammed Alfarraj, Rolla Mohammed Alshehri, Vicente Morales Bernal, Ali AlShaybani, Radi Ibrahim Alhamoud, Mohammed Ali Aljazeeri, Ling Kong Teng, Then Yee Ling, Daniel Thomas Melice

References

- Kelvin Abaa, John Wang, Derek Elsworth, Mku Ityokumbul. 2021. "Laboratory Investigation of Impact of Slickwater Composition on Multiphase Permeability Evolution in Tight Sandstones." *SPE* SPE 180250.
- Nancy Zakhour, Soodabeh Esmaili, Jonathan Ortiz, and Jack Deng. 2021. "High Viscosity Friction Reducer Testing, Trialing, and Application Workflow: A Permian Basin Case Study." *SPE* URTEC: 5249.
- P.S. Patel, PacWest Consulting Partners, PacWest Consulting Partners C.J. Robart, Kingsfield Consulting M. Ruegamer, and PacWest Consulting Partners A. Yang. 2014. "Analysis of US Hydraulic Fracturing Fluid System and Proppant Trends." *SPE* 168645.
- R.E. Hurst, J.M. Moore, Members Aime, D.E. Ramsey, and Member Aime Jr.. 1954. "DEVELOPMENT and APPLICATION of "FRAC" TREATMENTS in the PERMIAN BASIN." *SPE* T.P.4032.
- Christine Anderson, Adam Keilers, et al 2023. "Fit-for-Product Solution for Injecting Dry Friction Reducer During Stimulation Operations." *URTEC* 3989267.